

BREAST CANCER

Comprehensive Study Notes

DeVita, Hellman & Rosenberg's Cancer: Principles & Practice of Oncology

Chapter 106 – Malignant Tumors of the Breast

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1. EPIDEMIOLOGY & INTRODUCTION

Key Statistics (2009 US reference, Chapter basis):

Metric	Value
New breast cancer cases (USA)	192,370
Deaths from breast cancer (USA)	40,170
% of all female cancers	27%
% of female cancer deaths	15%
New European cases (2006)	429,900 (13.5% of all new cancers)
Decrease in US death rate since 1990	~24%

Drivers of mortality reduction (~equal contribution each):

- Adoption of screening mammography (earlier detection)
- Availability of adjuvant chemotherapy and tamoxifen

Global Trend:

- Historically less common in non-industrialized nations
- Incidence rising rapidly in developing countries with industrialization

2. ANATOMY OF THE BREAST

Component	Details
Location	Between 2nd and 6th ribs; sternal edge to mid-axillary line
Composition	Skin, subcutaneous tissue, breast tissue (epithelial 10–15% + stroma 85–90%)
Glandular lobes	15–20 lobes per breast; separated by adipose tissue
Blood supply	Internal mammary artery + lateral thoracic artery
Lymph drainage	>95% via axillary lymph nodes; <5% via internal mammary nodes
Internal mammary nodes	First 6 intercostal spaces, within 3 cm of sternal edge; highest concentration in spaces 1–3

Axillary Lymph Node Levels (by relation to pectoralis minor):

Level	Location	Clinical Significance
Level I	Lateral to pectoralis minor	First drainage; commonly involved first
Level II	Posterior to pectoralis minor	Skip metastases rare
Level III	Medial to pectoralis minor (infraclavicular)	Involved in high nodal burden; N3 disease

3. RISK FACTORS FOR BREAST CANCER

3.1 Genetic Risk Factors

BRCA1 & BRCA2 Gene Mutations:

Feature	BRCA1	BRCA2
Chromosome	17q21	13q12
Lifetime breast cancer risk	60–80%	40–85%
Lifetime ovarian cancer risk	39–46%	10–27%
Tumor phenotype	Triple-negative/Basal-like; high grade; medullary features	Similar to sporadic; excess tubular/lobular
ER/PR expression	Usually negative	Variable
HER2 overexpression	Infrequent	Infrequent
Associated cancers	Fallopian tube, prostate	Male breast, melanoma, gastric, prostate

BRCA Mutation Referral Criteria (USPSTF 2005)

- 2+ first-degree relatives with breast cancer diagnosed under age 50
- 3+ breast cancers at any age in family
- Ovarian cancer + breast cancer in same/different relatives
- BRCA-negative women with strong family history: still ~4× risk

Other Genetic Mutations:

Gene	Relative Risk	Notes
TP53 (Li-Fraumeni)	Very high	<1% of cases; multiple tumor types
PTEN (Cowden syndrome)	Very high	<1% of cases
CHEK2 (1100delC)	OR ~2.34	Low penetrance; 1.9% of cases; genetic testing premature
ATM heterozygote	~2×	Moderate risk

3.2 Hormonal Risk Factors

Factor	Effect on Risk	Mechanism
Early menarche	↑ (~20% decrease per year of delay)	Increased estrogen exposure
Late menopause	↑	Prolonged ovarian activity
Nulliparity / late first pregnancy (>30 y)	↑	Delayed hormonal differentiation of breast
Postmenopausal HRT (E+P)	↑ (especially combined)	Elevated plasma estrogen

Factor	Effect on Risk	Mechanism
Obesity (postmenopausal)	↑	Adipose aromatization of androgens → estrogens
Oophorectomy before age 50	↓ (magnitude ↑ with earlier age)	Estrogen deprivation; also protective in BRCA carriers
Breastfeeding	↓ slightly	Temporary ovarian suppression
Oral contraceptives	Slight ↑ (disappears after stopping)	Exogenous hormones

Key Hormonal Concept

Age-specific incidence increases steeply until menopause, then slows to ~1/6 of the premenopausal rate.

This confirms the critical role of ovarian estrogen in breast cancer development.

3.3 Benign Breast Disease

Category	Relative Risk	Examples
Non-proliferative	1× (no increase)	Cysts, mild hyperplasia, fibroadenoma without complex features
Proliferative without atypia	1.5–2.0×	Moderate/florid hyperplasia, sclerosing adenosis, papilloma
Proliferative with atypical hyperplasia (ADH/ALH)	4.0–5.0×	Atypical ductal or lobular hyperplasia
Atypia + 1st-degree family history of BC	11×	20% absolute risk at 15 years vs 8% without family history

3.4 Breast Density

- >75% mammographic density → 4.7× increased odds of breast cancer vs <10% density
- Density is partially genetically determined
- HRT increases density; discontinuation decreases it
- Dense breasts also reduce mammographic sensitivity

3.5 Environmental Risk Factors

Factor	Effect	Notes
Ionizing radiation	Strong ↑ (especially young age)	Mantle RT for Hodgkin lymphoma before age 15 → markedly elevated risk
Alcohol	↑ dose-dependent	~7% ↑ per drink/day
Physical inactivity	↑	Exercise is mildly protective
Electromagnetic fields	Uncertain	Inconsistent data
Organochlorine pesticides	Uncertain	No definitive evidence

4. SCREENING & PREVENTION

4.1 Screening Mammography

Guideline Body	Recommendation
American Cancer Society	Annual mammography beginning age 40 for average-risk women
USPSTF	Biennial ages 50–74; individualize 40–49
High-risk women ($\geq 20\%$ lifetime)	Annual mammography + annual breast MRI

- Meta-analysis: screening reduces relative risk of BC death by ~15% (RR 0.85, 95% CI 0.73–0.99)
- Mathematical models: screening + adjuvant therapy each contributed ~equally to the 24% mortality reduction

4.2 MRI Screening (American Cancer Society)

Recommendation Level	Indication
Annual MRI RECOMMENDED (evidence)	BRCA mutation carrier; untested 1st-degree relative of BRCA carrier; lifetime risk ≥ 20 –25%
Annual MRI RECOMMENDED (expert opinion)	Chest RT between ages 10–30 y; Li-Fraumeni syndrome; Cowden/Bannayan-Riley-Ruvalcaba (and 1st-degree relatives)
INSUFFICIENT evidence	Lifetime risk 15–20%; LCIS; atypical hyperplasia; extremely dense breasts; personal history of BC/DCIS

4.3 Risk Assessment Models

Model	Inputs	Limitations	Best Use
Gail Model	Age, menarche age, prior biopsies, atypia, 1st-degree relatives, race	Only 1st-degree relatives; no ovarian cancer; underestimates strong family history	Population risk estimation; validated in NSABP P1 (O:E ratio = 1.03)
Claus Model	1st- and 2nd-degree relatives with BC	Does not include non-family risk factors	Familial breast cancer risk estimation
BRCAPRO / Tyrer-Cuzick	Full pedigree + gene mutations	Requires software; complex	Predicting BRCA1/2 mutation probability

4.4 Chemoprevention

Tamoxifen Chemoprevention Trials:

Study	Follow-Up (mo)	Cancers	Placebo Rate	Tamox Rate	RR (95% CI)
Royal Marsden	70	70	5.0/1000 w-y	4.7	0.94 (0.59–1.43)
NSABP P-1	54.6	368	6.8	3.4	0.51 (0.39–0.66)
Italian	81.2	79	2.3	2.1	0.87 (0.62–2.14)
IBIS	96	337	6.8	5.0	0.73 (0.58–0.91)

- 4-trial overview: 38% reduction in breast cancer incidence ($P < 0.001$)
- 48% reduction in ER-positive cancers; NO effect on ER-negative cancers
- NSABP P-1: 49% risk reduction; 84% reduction in atypical hyperplasia subgroup
- NSABP P-2 (STAR): Raloxifene comparable efficacy to tamoxifen with fewer thromboembolic events

4.5 Prophylactic Surgery

Procedure	Population	Risk Reduction	Key Study
Bilateral prophylactic mastectomy	High-risk family history	90–94% reduction in BC incidence	Hartmann et al. (n=639, 14-year median FU)
Bilateral prophylactic mastectomy	BRCA1/2 carriers	90–100%	Meijers-Heijboer; Rebbeck et al.
Bilateral salpingo-oophorectomy (BSO)	BRCA1/2 carriers	51% ↓ BC risk (HR 0.49); 79% ↓ ovarian/fallopian (HR 0.21)	Kauff et al., Rebbeck meta-analysis

Clinical Pearl: BSO Dual Benefit

BSO reduces BOTH breast cancer risk AND ovarian cancer risk in BRCA carriers.

Ovarian cancer has no reliable screening — BSO eliminates this threat.

5. DIAGNOSIS & BIOPSY

MRI Diagnostic Limitation
Sensitivity: 88.1% Specificity: 67.7%
Abnormal MRI does NOT reliably confirm cancer.
Normal MRI does NOT reliably exclude cancer.
BIOPSY remains the diagnostic standard.

Biopsy Techniques Comparison:

Technique	Advantages	Disadvantages	Best Use
FNA (Fine-Needle Aspiration)	Rapid, painless, inexpensive, no incision	Cannot distinguish invasive vs DCIS; ER/PR/HER2 not available; needs cytopathologist; sensitivity 80–95%; FN rate 4–10%	Rapid prelim; palpable masses
Core-Cutting Needle Biopsy (CNB)	Rapid, any pathologist, ER/PR/HER2 available, no incision	Sampling error possible; FN occurs	PREFERRED for most lesions; essential pre-neoadjuvant therapy
Excisional Biopsy	Complete histology; may be definitive lumpectomy	Expensive, painful, incision, cosmetic deformity if benign	Reserved after failed/discordant CNB

Indications for Surgical Biopsy After Core Needle Biopsy:

- Failure to sample targeted calcifications
- Diagnosis of atypical ductal hyperplasia (ADH)
- Diagnosis of ALH or LCIS (controversial)
- Lack of concordance between imaging and histologic findings
- Radial scar (may mimic tubular carcinoma)
- Papillary lesions (cannot exclude papillary carcinoma in situ)

6. PATHOLOGY – IN SITU CARCINOMAS

6.1 Lobular Carcinoma In Situ (LCIS) / Lobular Neoplasia (LN)

LCIS/LN Definition

Proliferation of generally small, loosely cohesive cells originating in the terminal duct-lobular unit, with or without pagetoid involvement of ducts.

Includes: ALH (atypical lobular hyperplasia) and LCIS proper.

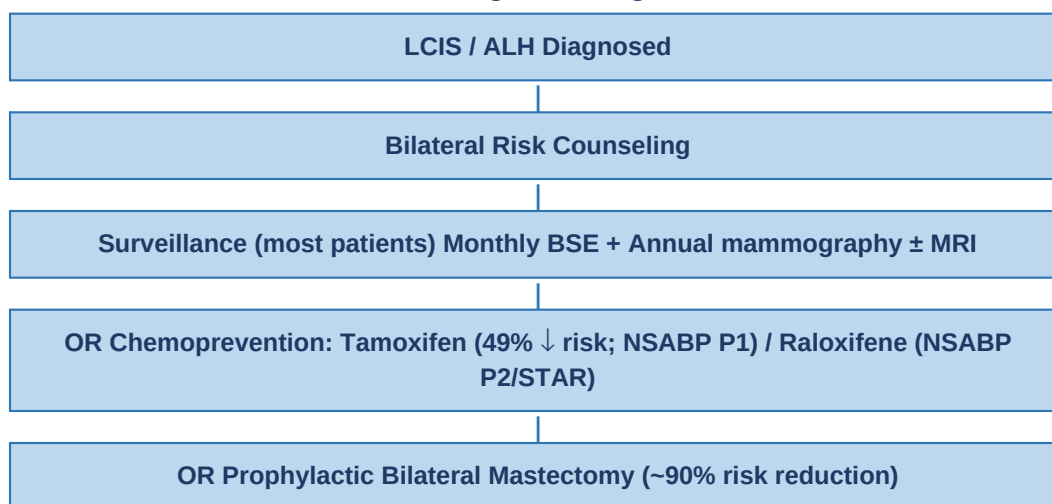
Feature	ALH	Classic LCIS	Pleomorphic LCIS
Risk of subsequent BC	Lower than LCIS	~10× RR; 30–40% lifetime	Possibly higher (data limited)
Laterality	Bilateral	Bilateral (ipsi > contra)	Bilateral
ER/PR	Positive	Positive	Positive
HER2	Negative	Negative	Negative
E-cadherin	Negative	Negative	Negative
Mammographic calcifications	Rare	Usually none	May be present (central necrosis)

Natural History (9-study analysis, n=172, ~10-year FU):

- 15% developed invasive carcinoma ipsilaterally; 9.3% contralaterally
- ~1–2% annual risk; 30–40% lifetime risk
- Subsequent cancer: more often IDC than ILC (ILC risk still substantially increased)

Management Algorithm:

LCIS Management Algorithm



- Negative margins NOT required for LCIS on excision
- Radiation therapy: NO established role for LN

6.2 Ductal Carcinoma In Situ (DCIS)

DCIS Definition

Proliferation of malignant-appearing mammary ductal epithelial cells

WITHOUT evidence of invasion beyond the basement membrane.

Epidemiology:

- Pre-mammography era: <5% of mammary cancers were DCIS
- Current screening era: 15–30% of screen-detected cancers are DCIS
- Greatest increase in women aged 49–69 years

Histologic Classification of DCIS:

Type	Features	Nuclear Grade
Comedo	Central necrosis; calcifications; high grade	High (Grade 3)
Cribriform	Punched-out spaces in solid sheets	Low-intermediate
Micropapillary	Finger-like projections into lumen	Low-intermediate
Papillary	Fibrovascular cores with epithelium	Low-intermediate
Solid	Solid sheets of cells	Variable

- 30–60% of DCIS lesions show mixed histologic patterns
- No single classification reliably predicts progression to invasive carcinoma

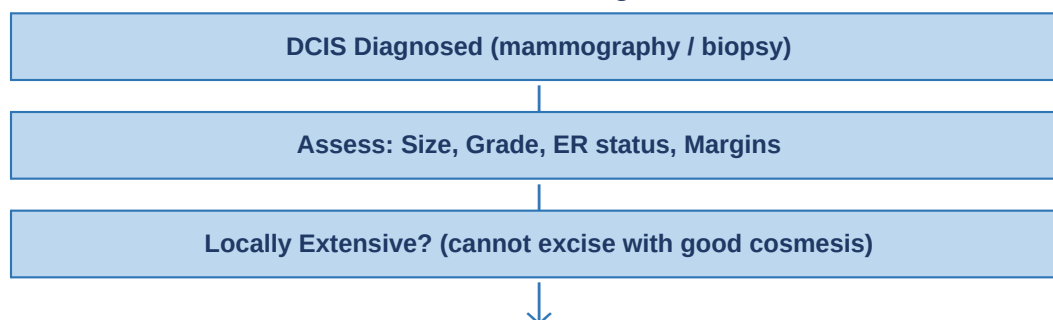
Randomized Trials: Excision ± Radiotherapy in DCIS:

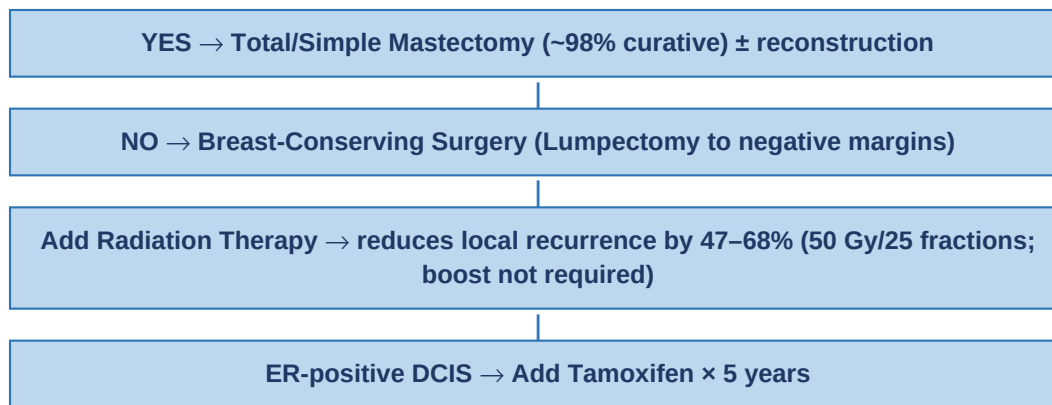
Trial (Result)	N	Recurrence WITHOUT RT	Recurrence WITH RT	Risk Reduction
NSABP B-17 (12-y)	813	31.7%	15.7%	50%
EORTC 10853 (10.5-y)	1,010	26%	15%	47%
UK/ANZ (12.7-y)	1,030	21%	8%	68%
Swedish (10-y)	1,067	29%	13%	60%

- ~50% of recurrences (with or without RT) are invasive carcinoma — not just DCIS
- No overall survival difference regardless of RT (cancer-specific survival >95%)
- Tamoxifen adds further reduction in local + contralateral events in ER+ DCIS

DCIS Treatment Algorithm:

DCIS Treatment Algorithm





7. PATHOLOGY – INVASIVE CARCINOMAS

Histologic Type	Frequency	Key Features	Prognosis
Invasive Ductal Carcinoma (IDC/NST)	70–80%	No special type; variable grade; most common	Baseline
Invasive Lobular Carcinoma (ILC)	10–15%	Indian file pattern; E-cadherin–; ER+; bilateral tendency; occult on mammogram	Similar or slightly better
Tubular Carcinoma	~2%	Well-differentiated tubules; ER+; usually node-negative	Excellent
Mucinous (Colloid)	~2%	Mucin pools; ER+; elderly women	Excellent
Medullary Carcinoma	~5%	Syncytial, lymphoid infiltrate, pushing margins; ER–/HER2–; BRCA1-associated	Favorable despite high grade
Inflammatory BC (IBC)	1–5%	Dermal lymphatic invasion; erythema/edema (peau d'orange); rapid onset; T4d	Poor — multimodality required

Paget Disease of the Breast

- ~1% of breast malignancies; in situ carcinoma of nipple epidermis
- Paget cells: large cells with clear cytoplasm and atypical nuclei in nipple epidermis
- Presentation: eczematoid changes, crusting, redness, erosion, nipple discharge/retraction
- May be associated with: (1) underlying invasive cancer; (2) underlying DCIS (Tis); (3) Paget alone (Tis)
- Treatment: BCT (central lumpectomy + RT) or mastectomy; SLNB if invasive cancer present

8. STAGING – AJCC TNM SYSTEM

8.1 T – Primary Tumor

T	Description
Tx	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	DCIS, LCIS, or Paget disease without invasive tumor
T1mi	Microinvasion ≤ 0.1 cm
T1a	Tumor 0.1–0.5 cm
T1b	Tumor 0.5–1.0 cm
T1c	Tumor 1.0–2.0 cm
T2	Tumor >2.0 cm to ≤ 5.0 cm
T3	Tumor >5.0 cm
T4a	Extension to chest wall (not pectoralis)
T4b	Skin involvement (edema, ulceration, satellite nodules)
T4c	Both T4a + T4b
T4d	Inflammatory breast cancer

8.2 N – Regional Lymph Nodes (Pathologic)

N	Description
N0	No regional lymph node metastasis
N0(i+)	ITC only (clusters ≤ 0.2 mm)
N1	1–3 axillary nodes positive; and/or internal mammary micrometastases on SLNB
N2	4–9 axillary nodes; or clinically detected internal mammary nodes
N3	≥ 10 axillary nodes; or infraclavicular; or supraclavicular nodes (previously M1)

8.3 Stage Groups

Stage	T	N	M	5-Year Survival
0	Tis	N0	M0	$>95\%$
IA	T1	N0	M0	$\sim 99\%$
IB	T0–1	N1mi	M0	$\sim 98\%$
IIA	T0–1 or T2	N1 or N0	M0	$\sim 85\text{--}90\%$
IIB	T2 or T3	N1 or N0	M0	$\sim 70\text{--}80\%$
IIIA	T0–3	N2	M0	$\sim 55\text{--}65\%$

Stage	T	N	M	5-Year Survival
IIIB	T4	N0–2	M0	~40–50%
IIIC	Any T	N3	M0	~40–50%
IV	Any T	Any N	M1	~20–25%

9. LOCAL-REGIONAL THERAPY: SURGERY & RADIATION

9.1 BCT vs Mastectomy

Feature	BCT (Lumpectomy + RT)	Mastectomy
Local recurrence (10-y)	~5–10%	~2–5%
Overall survival	Equivalent (NSABP B-06)	Equivalent
Breast preservation	Yes	No (reconstruction option)
Radiation required?	Yes (generally)	No (unless post-mastectomy RT indicated)

Absolute Contraindications to BCT:

- Multicentric disease (multiple quadrants)
- Tumor too large for adequate excision with acceptable cosmesis
- Positive margins after re-excision
- Diffuse malignant-appearing microcalcifications
- Prior breast/chest wall irradiation
- Pregnancy in 1st or 2nd trimester (radiation contraindicated)
- Patient preference for mastectomy

9.2 Radiation After BCT – EBCTCG Meta-Analysis

EBCTCG Key Finding (2005): n=42,000, 78 RCTs
For every 4 local recurrences prevented → 1 breast cancer death prevented.
RT after BCS: 15-year OS gain = 5.3% (P=0.005)
RT after mastectomy in node-positive: 15-year OS gain = 4.4% (P=0.001)
Benefit is similar regardless of whether local control achieved by surgery or RT.
RT benefit is amplified when adjuvant systemic therapy is used.

9.3 Post-Mastectomy Radiation (PMRT)

Indications for PMRT:

- ≥ 4 positive axillary lymph nodes: PMRT to chest wall + regional nodes (Level I evidence)
- 1–3 positive nodes: PMRT strongly recommended (growing evidence)
- T3/T4 tumors (locally advanced disease)
- Close (<1 mm) or positive margins after mastectomy

9.4 Axillary Surgery

Feature	Sentinel Lymph Node Biopsy (SLNB)	Axillary Lymph Node Dissection (ALND)
Indication	Clinically node-negative; primary BC	Clinically node-positive; high SLNB burden

Feature	Sentinel Lymph Node Biopsy (SLNB)	Axillary Lymph Node Dissection (ALND)
Lymphedema rate	~5–7%	~15–25%
5-year axillary failure (cN0, 1–3 positive sentinels)	0.9% (ACOSOG Z11)	0.5% (ACOSOG Z11)
Survival	Equivalent to ALND (Z11)	Standard for high nodal burden

- ACOSOG Z11: cN0 women with 1–3 positive sentinel nodes treated with BCT + systemic Rx + RT → ALND can be omitted without survival detriment
- IHC-only positive nodes (micrometastases ≤ 0.2 mm): routine ALND not required

9.5 Theories of Breast Cancer Spread

Theory	Concept	Implication
Halstedian	Strictly local disease; centrifugal spread	Radical surgery improves OS
Systemic (Fisher)	Systemic at diagnosis; predetermined ability to metastasize	Local therapy has no OS impact; surgery can be conservative
Spectrum (synthesis)	Window of opportunity before metastasis occurs	Both local AND systemic therapy affect OS

10. PROGNOSTIC & PREDICTIVE FACTORS

Definition Distinction

PROGNOSTIC: correlates with OS/DFS at diagnosis, independent of specific therapy.

PREDICTIVE: predicts response (or non-response) to a specific treatment.

Some factors are BOTH (e.g., ER/PR, HER2).

Factor	Type	Clinical Significance
Axillary lymph node status	Prognostic	Strongest established factor; stratify by 0, 1–3, 4–9, ≥10 nodes
Tumor size	Prognostic	Microscopic invasive size; larger = worse prognosis
Histologic grade (Nottingham)	Prognostic	Grade 1 → 3; use standardized methodology for reproducibility
ER/PR status	Both	ER+/PR+ → endocrine-sensitive; better 5-y DFS; difference less at 10 y
HER2 amplification/overexpression	Both	~20% of BC; predicts trastuzumab benefit; higher grade, poorer prognosis untreated
Lymphovascular invasion (LVI)	Prognostic	Independent adverse factor in node-neg and node-pos; requires careful IHC criteria
Patient age (<35 y)	Prognostic	Poorer prognosis; higher grade; less ER+; more LVI
Ki-67 / MIB-1	Prognostic	High = greater proliferation = worse outcome

HER2 Testing Criteria:

- IHC Positive: 3+ staining in >30% of invasive cancer cells
- FISH Positive: >6 copies/nucleus OR ratio >2.2
- IHC 2+ (equivocal): confirm with FISH
- ~20% gene amplification; ~5% overexpression without amplification

11. MOLECULAR SUBTYPES & GENOMIC TESTS

11.1 Intrinsic Molecular Subtypes (Sorlie/Perou)

Subtype	IHC Approximation	Prognosis	Key Clinical Points
Luminal A	ER+ and/or PR+, HER2–, low Ki-67	Favorable	Highly endocrine-sensitive; low chemo benefit; best prognosis overall
Luminal B	ER+ and/or PR+, HER2+ OR high Ki-67	Intermediate	Benefits from chemotherapy in addition to endocrine therapy
HER2-enriched	ER–, PR–, HER2+	Poor pre-trastuzumab; improved with targeted therapy	Trastuzumab-based therapy is essential
Basal-like/Triple Negative	ER–, PR–, HER2– (± CK5/6+, EGFR+)	Poor overall; high pCR to chemo	BRCA1-associated; no standard targeted therapy; immunotherapy emerging

11.2 Genomic Assays in Clinical Use

Test	Genes	Primary Use	Key Evidence
Oncotype DX (Recurrence Score)	21 genes (ER, PR, HER2, proliferation, invasion)	ER+/HER2–/node– (or 1–3 node+) BC; predict chemo benefit	RS <18 → omit chemo; RS >30 → add chemo (NSABP B-20, TAILORx)
MammaPrint	70-gene microarray	ER+/HER2–/node– BC: good vs poor prognosis	FDA-approved 2007; MINDACT trial validation
PAM50 (Prosigna)	50-gene RT-PCR	Intrinsic subtype + risk of recurrence score	Validated in postmenopausal ER+ node– and 1–3 node+ BC

Oncotype DX Clinical Decision Guide

Low RS (<18): Very low 10-year distant recurrence risk → NO benefit from chemotherapy

Intermediate RS (18–30): Most can omit chemo (premenopausal exception per TAILORx)

High RS (>30): High recurrence risk → SUBSTANTIAL benefit from chemotherapy

Advantage: Performed on paraffin-embedded tissue (no fresh tissue needed)

12. ADJUVANT SYSTEMIC THERAPY

Goal: Eradicate micrometastatic deposits present at diagnosis → prevent recurrence and improve survival.

12.1 Adjuvant Endocrine Therapy

Drug/Class	Mechanism	Indication	Duration	Key Benefit
Tamoxifen (SERM)	Competitive ER antagonist (partial uterus/bone agonist)	ER+ (pre- and postmenopausal)	5 y (extend to 10 y in high-risk)	~50% ↓ annual recurrence; ~28% ↓ annual mortality
Aromatase Inhibitors (AIs) anastrozole, letrozole, exemestane	Block aromatase → ↓ peripheral estrogen synthesis	Postmenopausal ER+ BC (or premenopausal with ovarian suppression)	5 y (upfront or sequential after tamoxifen)	Superior to tamoxifen in postmenopausal women; ↓ recurrence
GnRH agonists (goserelin, leuprolide)	Suppress ovarian estrogen production	High-risk premenopausal ER+ BC (combined with tamoxifen or AI)	2–5 y	SOFT/TEXT: combined with exemestane → ↑DFS vs tamoxifen alone in high-risk premenopausal

Tamoxifen Side Effects:

Side Effect	Relative Risk/Incidence	Notes
Endometrial cancer	2–3× increased	Monitor with annual pelvic exam
Thromboembolic events (DVT/PE)	~2× increased	Greater risk in immobile/older patients
Hot flashes / menopausal symptoms	Very common (~50%)	SSRIs can help (avoid paroxetine)
Bone density	Maintained/improved (postmenopausal); Lost (premenopausal)	Monitor bone density in premenopausal
Cataracts	Slight increase	Annual eye exam if symptomatic

12.2 Adjuvant Chemotherapy

Common Regimens:

Regimen	Drugs	Cycles	Key Indication
AC → T (dose-dense)	Doxorubicin + Cyclophosphamide → Paclitaxel	4+4 (q2w)	Node-positive; ER–; HER2+
TAC	Docetaxel + Doxorubicin + Cyclophosphamide	6×q3w	Node-positive
TC (no anthracycline)	Docetaxel + Cyclophosphamide	4×q3w	Low-risk; anthracycline-ineligible; ER+/HER2–

Regimen	Drugs	Cycles	Key Indication
FEC → D	5-FU + Epirubicin + Cyclophosphamide → Docetaxel	3+3	European standard
CMF	Cyclophosphamide + Methotrexate + 5-FU	6×	Low-risk; anthracycline-ineligible; older standard

12.3 HER2-Targeted Adjuvant Therapy

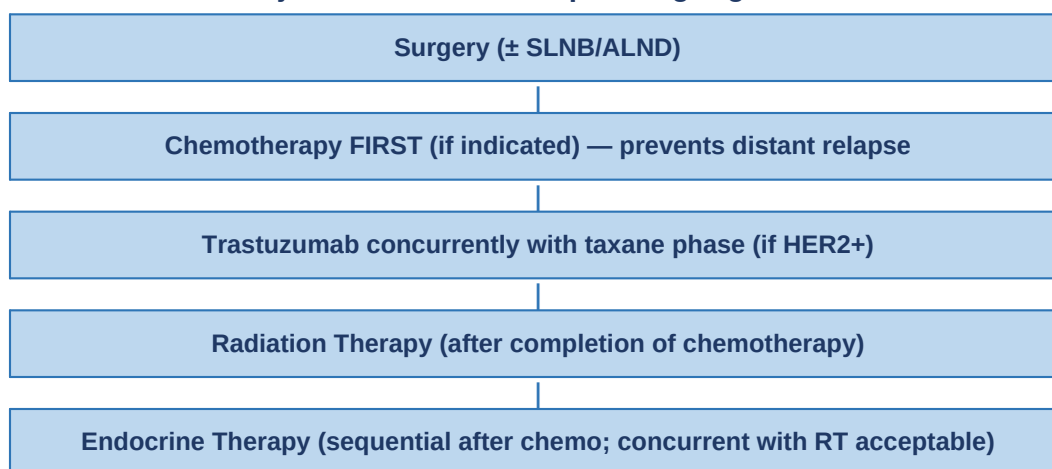
Trastuzumab (Herceptin) – Anti-HER2 Monoclonal Antibody:

Trial	N	Design	DFS Benefit	OS Benefit
NSABP B-31	2,043	AC→T ± trastuzumab (1y)	~52% ↓ recurrence	Significant (~33% ↓ mortality)
NCCTG N9831	3,505	AC→T ± concurrent/sequential trastuzumab (1y)	Concurrent best	Significant
HERA	5,090	Chemo → 1y vs 2y trastuzumab vs obs	~46% ↓ recurrence (1y)	Significant (1y = 2y)
BCIRG 006	3,222	AC→TH vs TCbH vs AC→T	TCbH ≈ AC→TH in DFS	TCbH: less cardiotoxicity

- Trastuzumab duration: 1 year is standard (2 years does not add benefit; HERA)
- Pertuzumab added to trastuzumab + chemo: improved DFS in node-positive HER2+ (APHINITY)
- T-DM1 (trastuzumab emtansine): adjuvant therapy for residual HER2+ disease post-neoadjuvant (KATHERINE)
- Neratinib: extended adjuvant after trastuzumab in high-risk HER2+/ER+ BC

12.4 Treatment Sequencing

Adjuvant Treatment Sequencing Algorithm

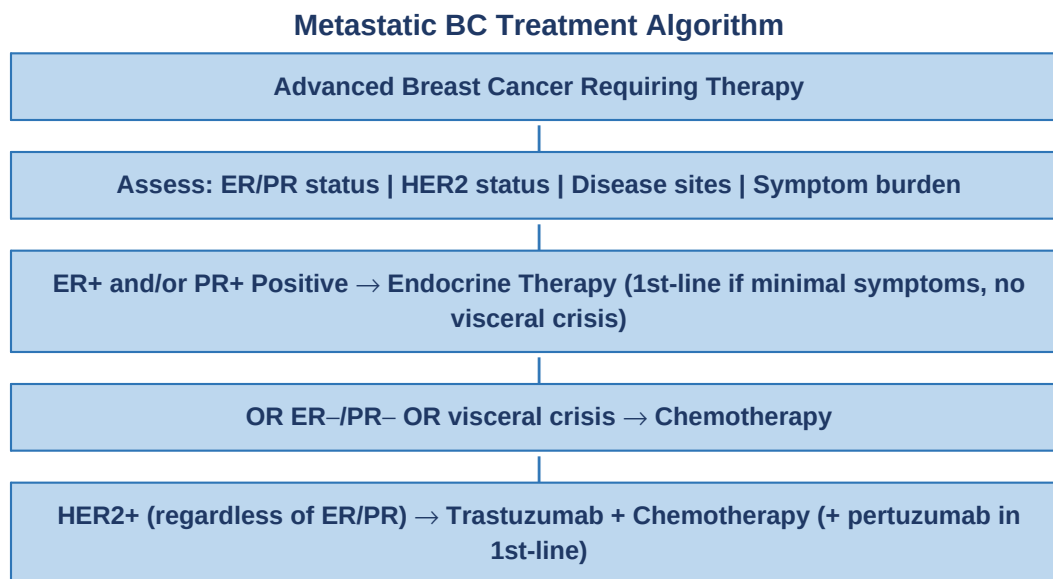


- Tamoxifen should NOT be given concurrently with chemotherapy (↑ recurrence risk)
- Concurrent AI + chemotherapy: by extrapolation, also probably best to avoid

13. METASTATIC (ADVANCED) BREAST CANCER

Goal: Prolongation of life, symptom control, quality of life maintenance. NOT generally curative.

13.1 Treatment Decision Algorithm



13.2 Endocrine Therapy for Metastatic BC

Agent	Class	Use	Approx Response	Duration
Tamoxifen	SERM	1st-line (pre- and post-menopausal)	~50% (naive ER+)	12–18 months
AIs (anastrozole, letrozole, exemestane)	Aromatase inhibitors	1st- or 2nd-line (postmenopausal)	~40–50%	8–12 months (1st-line)
Fulvestrant	SERD (pure anti-estrogen)	2nd-line; ± CDK4/6 inhibitor	~30–40%	4–6 months monotherapy
Megestrol acetate	Progestin	3rd-line and beyond	~25–30%	Variable
GnRH agonist + AI or tamoxifen	Ovarian suppression	Premenopausal metastatic ER+	Variable	Variable

- Tamoxifen flare: transient intensification of bone pain / hypercalcemia = indicates tumor response
- Resistance to endocrine therapy does NOT correlate with ER loss → multiple lines remain effective
- CDK4/6 inhibitors (palbociclib, ribociclib, abemaciclib) + endocrine therapy: markedly prolong PFS in ER+/HER2– metastatic BC

13.3 Chemotherapy for Metastatic BC

Category	Agents
Anthracyclines	Doxorubicin, Epirubicin, Pegylated liposomal doxorubicin (PLD)

Category	Agents
Taxanes	Paclitaxel (weekly preferred), Docetaxel, nab-Paclitaxel
Antimetabolites	5-FU (continuous infusion), Capecitabine (oral), Gemcitabine
Vinca Alkaloids	Vinorelbine, Vinblastine
Other	Ixabepilone (epothilone), Platinum salts, Etoposide, Cyclophosphamide
Common Combinations	AC, FAC, FEC, CMF; gemcitabine+paclitaxel; capecitabine+docetaxel; ixabepilone+capecitabine

13.4 HER2-Targeted Therapy (Metastatic)

Agent	Mechanism	Role in Metastatic BC	Key Data
Trastuzumab	Anti-HER2 mAb	1st-line + chemo; continue across multiple lines	Survival benefit; TTP improves with continuation beyond progression
Pertuzumab	Anti-HER2 mAb (dimerization domain)	1st-line + trastuzumab + docetaxel	CLEOPATRA: significant OS improvement
Lapatinib	Dual TKI (HER2 + EGFR)	2nd-line after trastuzumab + capecitabine	Improved TTP vs capecitabine alone; CNS activity
T-DM1 (ado-trastuzumab emtansine)	ADC: trastuzumab + DM1	2nd-line after trastuzumab + taxane	EMILIA: superior OS vs lapatinib + capecitabine

13.5 Bone Metastases

- Most common metastatic site in ER+ BC
- Bisphosphonates (zoledronic acid, pamidronate): reduce skeletal-related events (SREs)
- Denosumab (anti-RANKL): non-inferior or superior to zoledronic acid
- Palliative RT: highly effective for focal bone pain
- Surgical fixation: for impending/actual pathologic fractures

13.6 Brain Metastases

- More common in HER2+ and triple-negative BC subtypes
- Local therapy is primary: whole-brain RT, SRS (for limited lesions), surgery
- Systemic options with CNS activity: lapatinib (HER2+), capecitabine, some TKIs

14. SPECIAL THERAPEUTIC SITUATIONS

14.1 Locally Advanced BC (LABC) & Inflammatory BC (IBC)

Feature	LABC	Inflammatory BC (IBC)
AJCC Stage	IIIA–IIIC	IIIB (T4d)
Frequency	~15–20% of new cases	1–5% of all BC
Defining feature	Locally unresectable or bulky lymph nodes	Diffuse erythema/edema (peau d'orange); $\geq 1/3$ of breast skin; rapid onset
Key pathology	Variable	Dermal lymphatic tumor emboli (clinical diagnosis; not required pathologically)
Common error	—	Treated as mastitis → delayed diagnosis
Treatment	Neoadjuvant chemo → surgery → RT → adjuvant Rx	Neoadjuvant chemo FIRST (surgery contraindicated upfront) → mastectomy → RT → adjuvant Rx

14.2 Breast Cancer in Pregnancy

- Incidence: ~1 in 3,000 pregnancies; most common cancer diagnosed in pregnancy
- Diagnosis often delayed due to breast changes masking the lesion
- Surgery: safe throughout pregnancy (modified radical mastectomy or BCT feasible)
- Chemotherapy SAFE in 2nd/3rd trimester: FAC (5-FU, doxorubicin, cyclophosphamide) — no complications reported
- AVOID in pregnancy: Methotrexate (abortion/fetal malformation); Tamoxifen (uncertain); Trastuzumab (reversible anhydramnios)
- Radiation therapy: contraindicated during pregnancy; give post-delivery
- Breastfeeding: avoid while receiving chemotherapy or tamoxifen

14.3 Male Breast Cancer

Feature	Details
Incidence (USA 2007)	~2,030 cases; ~450 deaths; 1.1% of all BC
Risk factors	Elevated estrogen (Klinefelter, liver disease), reduced androgen, BRCA2 (>BRCA1), radiation
Histology	IDC most common; ~80–90% ER-positive
Treatment	Follows same principles as female BC; tamoxifen standard endocrine therapy; AIs less effective
Prognosis	Stage-for-stage similar; often diagnosed at later stage

14.4 Occult Primary Breast Cancer

- Axillary lymphadenopathy without identifiable breast primary on physical exam or mammography
- ER/PR/HER2 IHC on lymph node strongly suggests breast origin

- MRI identifies primary in ~62% of cases with normal mammography
- If primary found: treat with standard BCT or mastectomy
- If no primary found: mastectomy or whole-breast RT (acceptable alternative)

15. FOLLOW-UP GUIDELINES (ASCO)

Recommended Surveillance

Test	Frequency	Notes
History & Physical Exam	Q3–6 months (years 1–3); Q6–12 months (years 4–5); annually thereafter	Look for local/regional/contralateral recurrence; treatment side effects
Mammography	Annually (starting ≥6 months after RT)	Both breasts (BCT); contralateral (mastectomy)
Breast Self-Examination	Monthly	Supplement to clinical exam
Pelvic Examination	Annually	Tamoxifen users: monitor for endometrial changes

NOT Recommended for Routine Surveillance

Test NOT Recommended	Reason
CBC / Liver function tests	Not cost-effective in asymptomatic patients
Chest X-ray, Bone scan	Low yield without symptoms/signs
CT / PET-CT / FDG-PET	Not for routine surveillance; use for staging/restaging
Breast MRI (routine)	Not recommended unless high-risk criteria met
Tumor markers (CA 15-3, CA 27.29, CEA)	Not recommended for routine surveillance

16. QUICK REFERENCE TABLES

16.1 Risk Factor Relative Risks Summary

Risk Factor	Relative Risk	Notes
BRCA1 mutation	60–80% lifetime risk	High penetrance
BRCA2 mutation	40–85% lifetime risk	High penetrance
Atypical hyperplasia + family history	11×	Highest non-genetic RR
Atypical hyperplasia alone	4–5×	Without family history
LCIS	~10× (30–40% lifetime)	Bilateral risk marker
Proliferative disease without atypia	1.5–2×	Modest increase
Mammographic density >75%	4.7× odds	vs <10% density
Mantle RT (Hodgkin, <15 y)	Very high (>20×)	MRI screening recommended
Postmenopausal HRT (E+P)	~1.5–2×	Disappears after stopping

16.2 Tamoxifen vs Aromatase Inhibitors in Adjuvant Setting

Feature	Tamoxifen	Aromatase Inhibitors
Mechanism	Competitive ER antagonist (partial agonist uterus/bone)	Block aromatase → ↓ peripheral estrogen
Menopausal status	Pre- AND postmenopausal	Postmenopausal ONLY (or with ovarian suppression)
Efficacy	Standard (5-year)	Superior DFS in postmenopausal women
Endometrial cancer risk	↑ 2–3×	Not increased (possibly ↓ vs tamoxifen)
Thromboembolism	↑ ~2×	Not significantly increased
Bone density	Maintained/improved (postmenopausal); Lost (premenopausal)	↓ Bone density → osteoporosis risk
Arthralgias/myalgias	Less common	Common (joint/muscle pain — class effect)

16.3 DCIS vs LCIS Key Differences

Feature	DCIS	LCIS/ALH
Origin	Ductal epithelium	Lobular epithelium (TDLU)
E-cadherin	Positive	Negative
Mammographic finding	Microcalcifications (most common)	Usually occult; no specific imaging feature

Feature	DCIS	LCIS/ALH
Laterality of risk	Ipsilateral (same breast/quadrant)	Bilateral
Margin requirement	Negative margins required for BCT	NOT required on excision
Radiation role	Reduces local recurrence by 47–68%	No established role

16.4 BC Subtype: Key Treatment Summary

Subtype (IHC)	Key Systemic Therapy	Avoid	Chemo Benefit
ER+ and/or PR+, HER2– (Luminal A/B)	Endocrine therapy ± chemo based on Oncotype DX RS / risk features	Trastuzumab (HER2–)	Oncotype DX RS guides chemo decision
ER–, PR–, HER2+ (HER2-enriched)	Trastuzumab + chemo (AC→TH or TCbH); + pertuzumab in high-risk	Endocrine therapy alone	High — anthracycline/taxane-based
ER+ and/or PR+, HER2+ (Luminal B HER2+)	Endocrine therapy + trastuzumab + chemo	—	High — dual HER2 blockade
ER–, PR–, HER2– (Triple Negative)	Chemo (anthracycline/taxane) ± pembrolizumab ± PARP inhibitor (BRCA-mutated)	Endocrine therapy; trastuzumab	High pCR rate to neoadjuvant chemo

— END OF STUDY NOTES —

Source: DeVita, Hellman & Rosenberg's Cancer: Principles & Practice of Oncology

Chapter 106 – Malignant Tumors of the Breast

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